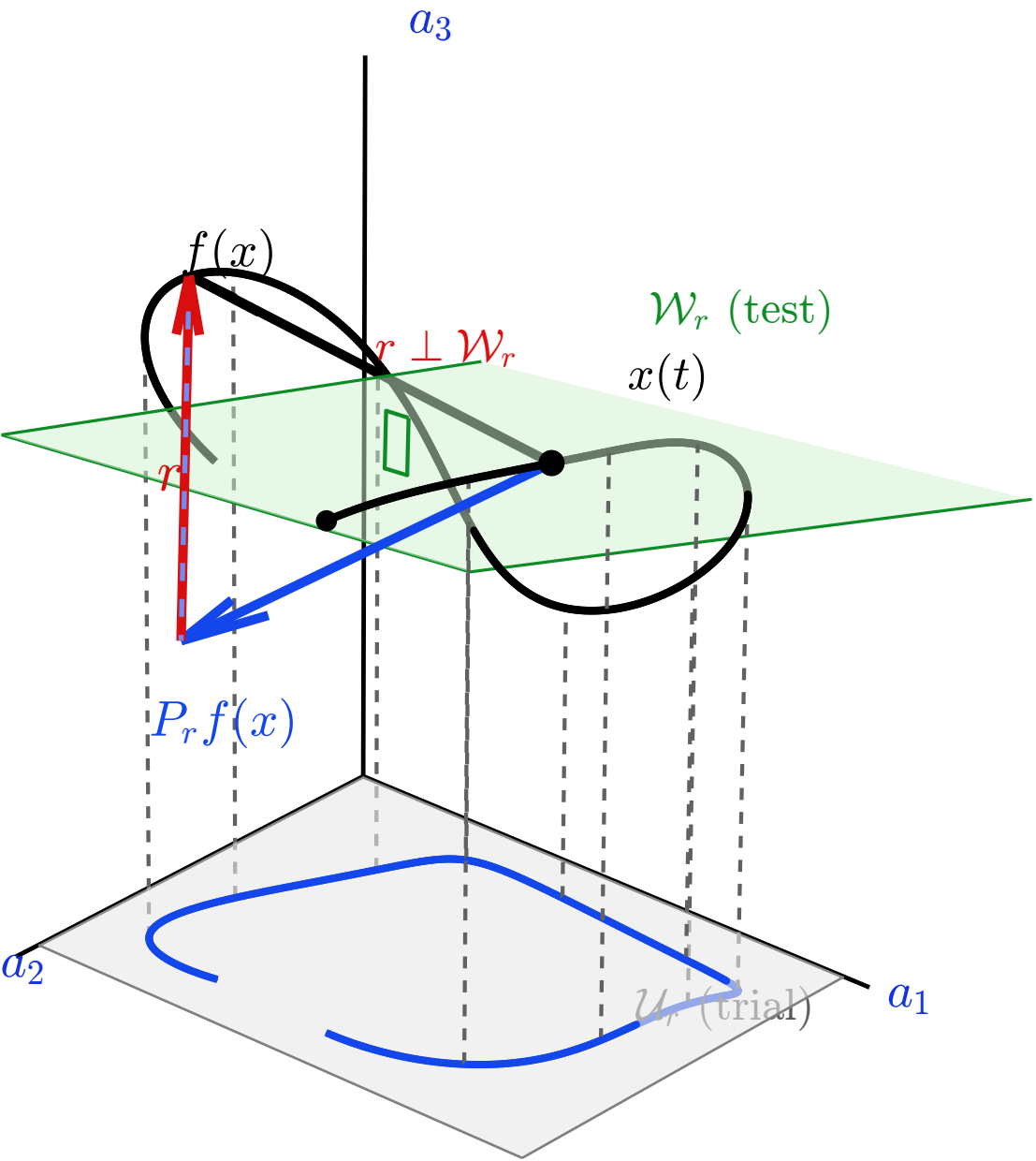

Petrov-Galerkin condition

$\mathcal{W}_r \neq \mathcal{U}_r$

$P_r = U_r(W_r^\top U_r)^{-1} W_r^\top$

$r = f(x) - P_r f(x)$

$W_r^\top r = 0$

$r \perp \mathcal{W}_r$ (not $\perp \mathcal{U}_r$)